0.1 Conditional (Ternary) Operator (`?:`) Practice Questions in C

Note: Use only the **conditional operator (`?:`)** wherever possible. Avoid `if-else` unless specifically required.

0.1.1 Beginner Level

1. Print the larger of two numbers.
   - **Hint:** `a > b ? a : b`
2. Print the smaller of two numbers.
   - **Hint:** Reverse the comparison.
3. Check whether a number is positive or negative.
   - **Hint:** Condition should return a string.
4. Check whether a number is even or odd.
   - **Hint:** Use `% 2`.
5. Find the absolute value of a number.
   - **Hint:** Negative numbers need sign change.
6. Print "Pass" if marks 35, otherwise "Fail".
   - **Hint:** Simple comparison.
7. Find the maximum of three numbers.
   - **Hint:** Use nested ternary operators.
8. Find the minimum of three numbers.
   - **Hint:** Similar to maximum.
9. Check whether a character is uppercase or lowercase.
   - **Hint:** Compare with `'A'` and `'Z'`.
10. Check whether a number is zero or non-zero.

- **Hint:** Direct condition.

0.1.2 Easy-Intermediate Level

11. Determine whether a year is a leap year or not.
    - **Hint:** Use leap year formula inside ternary.
12. Find whether a number is divisible by 5.
    - **Hint:** `% 5 == 0`
13. Check whether a number is divisible by both 3 and 7.
    - **Hint:** Use `&&`.
14. Print the sign of a number (+, -, 0).
    - **Hint:** Nested ternary.
15. Find the greatest among four numbers.
    - **Hint:** Compare pairwise.
16. Find the smallest among four numbers.
    - **Hint:** Similar approach.
17. Check whether a character is a vowel or consonant.
    - **Hint:** Compare against vowels.
18. Find whether a character is an alphabet or not.
    - **Hint:** Check uppercase and lowercase ranges.
19. Check whether a number lies between 10 and 50.
    - **Hint:** Use `&&`.
20. Print “Eligible” if age 18 else “Not Eligible”.
    - **Hint:** Single condition.

0.1.3 Intermediate Level

21. Find the middle value among three numbers.
    - **Hint:** Multiple comparisons.
22. Check whether a number is a two-digit number.
    - **Hint:** Absolute value between 10 and 99.
23. Find the larger of two floating-point numbers.
    - **Hint:** Use ternary with `float`.

24. Find the largest digit in a two-digit number.
    - **Hint:** Extract digits first.
25. Check whether a character is a digit.
    - **Hint:** `'0' <= ch <= '9'`
26. Convert uppercase to lowercase using ternary.
    - **Hint:** ASCII difference.
27. Convert lowercase to uppercase using ternary.
    - **Hint:** ASCII difference.
28. Print grade:
    - Marks $90 \rightarrow$ A
    - Marks $75 \rightarrow$ B
    - Marks $60 \rightarrow$ C
    - Otherwise D
    - **Hint:** Nested ternary chain.
29. Check whether a number is a multiple of 10.
    - **Hint:** `% 10 == 0`
30. Find the nearest greater value between two numbers.
    - **Hint:** Compare directly.

0.1.4 Advanced Level

31. Find the maximum of five numbers.
    - **Hint:** Nested comparisons.
32. Find the minimum of five numbers.
    - **Hint:** Similar logic.
33. Determine whether a character is:
    - Uppercase
    - Lowercase
    - Digit
    - Special Character
    - **Hint:** Multiple nested ternaries.
34. Classify a temperature:
    - $< 0 \rightarrow$ Freezing
    - 0–20 $\rightarrow$ Cold
    - 21–35 $\rightarrow$ Warm

- 35 → Hot
- **Hint:** Nested ternary ladder.

35. Determine salary bonus:
    - Salary > 50000 → 20%
    - Salary > 30000 → 10%
    - Otherwise 5%
    - **Hint:** Nested ternary.
36. Find the second largest among three numbers.
    - **Hint:** Exclude largest first.
37. Find the second smallest among three numbers.
    - **Hint:** Similar logic.
38. Determine if a triangle is valid using three sides.
    - **Hint:** Triangle inequality.
39. Determine the largest among three entered characters (ASCII comparison).
    - **Hint:** Character comparisons.
40. Check whether a number belongs to:
    - Negative
    - Zero
    - Positive
    - **Hint:** Three-way nested ternary.

0.1.5 Challenge Problems

41. Find the median of three numbers.
    - **Hint:** Use only ternary operators.
42. Check whether a character is a hexadecimal digit.
    - **Hint:** 0–9, A–F, a–f.
43. Find the maximum of six numbers without loops.
    - **Hint:** Repeated nesting.
44. Assign grades with distinction:
    - 90 Distinction
    - 75 First Class
    - 60 Second Class
    - 35 Pass
    - Otherwise Fail
    - **Hint:** Deep ternary chain.

45. Determine whether a given time (hour) is:
    - Morning
    - Afternoon
    - Evening
    - Night
    - **Hint:** Range-based nesting.
46. Check whether a point `(x,y)` lies in:
    - Quadrant I
    - Quadrant II
    - Quadrant III
    - Quadrant IV
    - Axis
    - **Hint:** Nested ternary.
47. Find the largest among seven numbers.
    - **Hint:** Build step-by-step maximums.
48. Determine whether a character is:
    - Alphabet
    - Digit
    - Special Symbol
    - **Hint:** Combine range checks.
49. Print "Adult Male", "Adult Female", "Boy", or "Girl".
    - **Hint:** Use age and gender inputs.
50. Build a simple calculator using only conditional operators for:
    - Addition
    - Subtraction
    - Multiplication
    - Division
    - **Hint:** Operator character decides the expression.

0.1.6 Expert-Level Nested Ternary Challenges

51. Largest of 10 numbers.
52. Smallest of 10 numbers.
53. Rank three students based on marks.
54. Check whether a date is valid.
55. Determine triangle type (Equilateral, Isosceles, Scalene).
56. Determine angle type (Acute, Right, Obtuse).
57. Check voting eligibility based on age and citizenship status.
58. Find second largest among five numbers.
59. Find third largest among five numbers.
60. Implement menu selection using only the conditional operator.